

JIAYI KANG

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💡 RESEARCH INTERESTS

Fully Homomorphic Encryption (FHE): theoretical foundations (e.g. bootstrapping) and privacy-preserving applications (e.g. secure machine learning, private information retrieval).

Zero-knowledge proofs (ZKP): with an emphasis on post-quantum constructions (e.g. lattice-based ZKP).

Intersection of FHE and ZKP: secure verifiable cryptographic protocols that provide both privacy and integrity.

🎓 EDUCATION

PhD in Cryptography 2021 - 2025

KU Leuven, Department of Electrical Engineering, COSIC research group

Thesis: Optimisations and applications of fully homomorphic encryption 📄

Supervised by Prof. Frederik Vercauteren, Prof. Nigel Smart and Dr. Ilia Iliashenko

MSc in Mathematics (*With Great Distinction*) 2019 - 2021

KU Leuven, Department of Mathematics

Master of Physics (*First Class Honors*) 2015 - 2017

The University of Manchester, Department of Physics and Astronomy

BSc in Physics (*Honor Science Program*) 2012 - 2016

Xi'an JiaoTong University, Department of Physics

Exchange to University of California, Berkeley in 2015 Spring

🏢 EXPERIENCE

PostDoc Researcher | KU Leuven Dec. 2025- Present

Research on fully homomorphic encryption and lattice-based cryptographic protocols, with a focus on their interaction and applications, under the supervision of Prof. Frederik Vercauteren.

Research Visit | Seoul National University Jul.-Aug. 2023

Visiting researcher in the Cryptography Group led by Prof. Jung Hee Cheon. The visit was supported by the Industrial & Mathematical Data Analytics Research Center (IMDARC), Seoul National University, Korea.

Research Intern | Intel Labs Jul. - Sep. 2022

Privacy Technologies Graduate Research Intern at Intel Labs, hosted by Dr. Rosario Cammarota and supervised by Dr. Charlotte Bonte and Dr. Duhyeong Kim.

Research Assistant | The Chinese University of Hong Kong 2017-2019

Research assistant in the Department of Physics

⚙️ PUBLICATIONS

(authors ordered alphabetically except for publications marked with *)

Conferences

- Jacob Blindenbach, Jung Hee Cheon, Gamze Gürsoy, **Jiayi Kang**. On the overflow and p -adic theory applied to Homomorphic Encryption, in *Cyber Security, Cryptology, and Machine Learning (CSCML)* 2024 📄
- Kelong Cong, **Jiayi Kang**, Georgio Nicolas, Jeongeun Park. Faster Private Decision Tree Evaluation for Batched Input from Homomorphic Encryption, in *Security and Cryptography for Networks (SCN)* 2024 📄
- Kelong Cong, Robin Geelen, **Jiayi Kang**, Jeongeun Park. Revisiting Oblivious Top- k Selection with Applications to Secure k -NN Classification, accepted in *Selected Areas in Cryptography (SAC)* 2024 📄
- Robin Geelen, Ilia Iliashenko, **Jiayi Kang**, Frederik Vercauteren. On Polynomial Functions Modulo p^e and Faster Bootstrapping for Homomorphic Encryption, in *EUROCRYPT* 2023 📄

Journals

- **Jiayi Kang**, Leonard Schild. Pirouette: Query Efficient Single-Server PIR, to appear in *Proceedings on Privacy Enhancing Technologies (PoPETs)* 2026 
- * Hua Xu¹, Mariana Gama, Emad Heydari Beni, **Jiayi Kang**. FRItata: A FRI-based Polynomial Commitment Scheme for Distributed Proof Generation, *Communications in Cryptology (CiC)* 2025 
- Mariana Gama, Emad Heydari Beni, **Jiayi Kang**, Jannik Spiessens, Frederik Vercauteren. Blind zkSNARKs for Private Proof Delegation and Verifiable Computation over Encrypted Data, *Communications in Cryptology (CiC)* 2025 
- * Jacob Blindenbach¹, **Jiayi Kang**¹, Seungwan Hong¹, Caline Karam, Thomas Lehner, and Gamze Gürsoy. Ultra-secure storage and analysis of genetic data for the advancement of precision medicine, in *Genome Biology* 2024 

TALKS AND SEMINARS

- Blind zkSNARKs for Private Proof Delegation and Verifiable Computation over Encrypted Data, invited seminar at IIE Chinese Academy of Sciences, Beijing, China, 2025
- Blind zkSNARKs for Private Proof Delegation and Verifiable Computation over Encrypted Data at *the 4th Annual FHE.org Conference on Fully Homomorphic Encryption*, Sofia, Bulgaria, 2025 
- On the overflow and p -adic theory applied to Homomorphic Encryption at *Cyber Security, Cryptology, and Machine Learning (CSCML)*, virtual, 2024
- Faster Private Decision Tree Evaluation for Batched Input from Homomorphic Encryption at *Security and Cryptography for Networks (SCN)*, Amalfi, Italy, 2024
- Revisiting Oblivious Top- k Selection with Applications to Secure k -NN Classification, invited seminar at University of Luxembourg, 2024
- On Polynomial Functions Modulo p^e and Faster Bootstrapping for Homomorphic Encryption at *the 2nd Annual FHE.org Conference on Fully Homomorphic Encryption*, Tokyo, Japan, 2023 

GRANTS AND AWARDS

- Private AI Visiting Scholarship | IMDARC, Korea 2023
- Doctoral Scholarship | KU Leuven, Belgium 2021-2025

TEACHING

- Guest Lecturer for the KU Leuven course *Advanced Methods in Cryptography* (2026 Spring)
- Invited Lecturer for the program *Rings and Finite Fields in Modern Cryptography* at the Advanced School for Computing and Imaging (ASCI), the Netherlands (2026)
- Guest Lecturer for the KU Leuven course *Privacy and Big Data* (2023 Fall, 2024 Fall, 2025 Fall)
- Guest Lecturer for the KU Leuven course *Privacy Technologies* (2024 Fall)
- Teaching Assistant for the KU Leuven course *Computer Algebra for Cryptography* (2023 Spring, 2024 Spring)

MASTER THESIS CO-SUPERVISON

- Fernando Javier Lopez Cerezo (2025-Present), Lattice-based anonymous credential
- Antoine Janssens (2025-Present), Private information retrieval with GBFV: analysis and implementation
- Hua Xu (2024-2025), Distributed Proof Generation of FRI-based SNARKs
- Sabine Chentouf (2024-2025), Lattice-based zero-knowledge proofs for privacy-preserving federated learning
- Pritam Pal (2023-2024), From zero to HEro: zkSNARKs Proof generation with Homomorphic Encryption
- Yingshuo Xi (2022-2023), An Investigation of Polynomial Activation Functions in Neural Networks
- Siva Kumar (2022), Secure Data Classification with Homomorphic Encryption

COMMUNITY SERVICES

Program committee

- ACM CCS 2026
- The FHE.org conference 2026
- ACM Web conference (WWW) 2026

Reviewer

- Journal of Cryptology (JoC) 2026
- Designs, Codes and Cryptography (DCC) 2024

External reviewer

- PKC 2026, USENIX 2026, CRYPTO 2026
- Eurocrypt 2025, Asiacrypt 2025
- Eurocrypt 2024, WAHC 2024
- Asiacrypt 2023, CHES 2023